

Assignment No.1 2413FEB1T1 : Engineering Mathematics-1

Topic: Matrices

1. Prove that every square matrix can be uniquely expressed as the sum of symmetric and skew-symmetric matrices.
2. Express the following matrix as the sum of a symmetric and a skew-symmetric matrix

$$A = \begin{bmatrix} 3 & -2 & 6 \\ 2 & 7 & -1 \\ 5 & 4 & 0 \end{bmatrix}$$

3. Express the matrix as a sum of the Hermitian and skew-Hermitian matrix.

$$\begin{bmatrix} 3i & -1+i & 3-2i \\ 1+i & -i & 1+2i \\ 3-2i & -1+2i & 0 \end{bmatrix}$$

4. Verify that the following matrices are orthogonal and hence find their inverse.

(a)

$$A = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$$

(b)

$$B = \begin{bmatrix} \sin x & -\cos x & 0 \\ \cos x & \sin x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Find a, b and c if A is an orthogonal matrix. Hence find the inverse of A.

$$A = \frac{1}{3} \begin{bmatrix} a & b & c \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$$

6. Prove that the following matrix is orthogonal and hence find the inverse of A.

$$A = \frac{1}{3} \begin{bmatrix} -2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{bmatrix}$$

7. Prove that the matrix A is unitary and hence find inverse of A where

$$A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$$

8. Determine the values of k such that the rank of the A is 3 where

$$A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ k & 2 & 2 & 2 \\ 9 & 9 & k & 3 \end{bmatrix}$$

9. Find the rank of the following matrix by reducing it to echelon form.

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$$

10. Find the rank of the following matrix by reducing it to normal form.

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

11. Find the rank of the following matrix by reducing it to normal form

$$\begin{bmatrix} 2 & -1 & 1 & 1 \\ 1 & 0 & 1 & 2 \\ 3 & 3 & 3 & 1 \\ 1 & 4 & 2 & 0 \\ 0 & -4 & -1 & 2 \end{bmatrix}$$

12. Find non-singular matrices P and Q such that PAQ is in normal form and hence find the rank of A for the following matrices.

(a)

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & 1 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 0 & -2 \\ 2 & 3 & -4 \\ 3 & 3 & -6 \end{bmatrix}$$

13. Find non-singular matrices P and Q such that PAQ is in normal form and hence find the rank of A for the following matrix

$$\begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 1 & 4 & -5 \\ -1 & -5 & -5 & 7 \end{bmatrix}$$

14. Solve the equation

$$2x - 2y - 5z = 0$$

$$4x - y + z = 0$$

$$3x - 2y + 3z = 0$$

$$x - 3y + 7z = 0$$

15. For what values of λ the system of equations have non-trivial solution? Obtain the solution for real values of λ where

$$3x + y - \lambda z = 0$$

$$4x - 2y - 3z = 0$$

$$2\lambda x + 4y + \lambda z = 0$$

16. Test for consistency the following systems and solve them if consistency:

$$x - 2y + 6z = 8$$

$$x + 2y - 3z = -1$$

$$15x - 3y + 9z = 21$$

17. Investigate for what values of λ and μ , the equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

(a) no solution

(b) a unique solution

(c) infinite no. of solution

18. Encode the message 'SECRET CODE' with the key matrix

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 6 \end{bmatrix}$$

19. Decode the message as 7,7,4,16,5,8,9,13,20,3, with the Hill 2-cipher key

$$A = \begin{bmatrix} 1 & 0 \\ 4 & 13 \end{bmatrix}$$

20. Use the matrix

$$A = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix}$$

to decode the message matrix

$$C = \begin{bmatrix} 4 & 11 & 12 & -2 \\ -4 & 4 & 9 & -2 \end{bmatrix}$$

Subject: 2413FEB1T1 : Engineering Mathematics-1
Assignment No.3 Topic: Successive Differentiation

Q.1 Find the n^{th} derivative of $y = \frac{1}{x}$

Q.2 If $y = \frac{1}{x^2 - 5x + 6}$ then find y_n

Q.3 If $y = \frac{x}{(x-1)(x-2)(x-3)}$ then find y_n

Q.4 Find the n^{th} derivative of the function $y = \sin x \sin 2x \sin 3x$

Q.5 Find the n^{th} derivative of $e^x \sin 2x \cos 3x$

Q.6 Find the n^{th} derivative of $\frac{x^3}{(x+1)(x-2)}$

Q.7 Find the n^{th} derivative of $x^3 \cos x$

Q.8 If $y = \frac{\log x}{x}$ then prove that $y_5 = \frac{5!}{x^6} \left[1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \log x \right]$

Q.9 If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$ then prove that $x^2 y_{n+2} + (2n+1)xy_{n+1} + 2n^2 y_n = 0$

Q.10 If $y^{1/m} + y^{-1/m} = 2x$ show that $(x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$

Q.11 If $y = \cos(m \sin^{-1} x)$ then prove that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$

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Assignment No.2 Topic: Partial Differentiation and Applications

1. If $u = x^y$ then show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$
2. If $u = \log(x^2 + y^2)$ then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$
3. If $u = (1 - 2xy + y^2)^{-\frac{1}{2}}$ then prove that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3$
4. If $u = e^{xyz}$ then prove that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2 y^2 z^2) e^{xyz}$
5. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that
 - (a) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$
 - (b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + 2 \frac{\partial^2 u}{\partial y \partial z} + 2 \frac{\partial^2 u}{\partial x \partial z} = \frac{-9}{(x+y+z)^2}$
6. If $z = f(x, y)$ where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$ then show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$
7. If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$
8. If $u = f(x^2 - y^2, y^2 - z^2, z^2 - x^2)$ then show that $\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} + \frac{1}{z} \frac{\partial u}{\partial z} = 0$
9. If $z = \tan^{-1}(\frac{y}{x})$ then show that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$
10. Find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ where $u = \frac{\sqrt{x} + \sqrt{y}}{x+y}$.
11. If $f(x, y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2}$ show that $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + 2f(x, y) = 0$
12. If $u = \tan^{-1}(\frac{x^3 + y^3}{x - y})$, $x \neq y$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$
13. If $u = \log(\sqrt{x^2 + y^2})$ then prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + 1 = 0$
14. If $u = \tan^{-1}(\frac{x^3 + y^3}{2x + 3y})$ then prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = (1 - 4\sin^2 u) \sin 2u$
15. Examine the function $f(x, y) = xy(3 - x - y)$ for extreme values and also find maximum and minimum values of $f(x, y)$
16. Find the maximum and minimum value of $x^3 + 3xy^2 - 3x^2 + 4$.
17. Find the stationary values of $x^3 + y^3 - 3axy$; $a > 0$

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Assignment No.4 Topic: Indeterminate Forms

1. Evaluate $\lim_{x \rightarrow 0} \frac{\log(1+kx^2)}{1-\cos x}$
2. Evaluate $\lim_{x \rightarrow 0} \frac{a^x - 1 - x \log a}{x^2}$
3. Evaluate $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$
4. Find values of a, b if $\lim_{x \rightarrow 0} \frac{a \cos x - a + bx^2}{x^4} = \frac{1}{12}$
5. Find values of a, b if $\lim_{x \rightarrow 0} \frac{a \sinh x + b \sin x}{x^3} = \frac{5}{3}$
6. Find values of a, b if $\lim_{x \rightarrow 0} \frac{a \sin^2 x + b \log(\cos x)}{x^4} = -\frac{1}{2}$
7. Evaluate $\lim_{x \rightarrow 0} \log_{\tan x} \tan 2x$
8. Evaluate $\lim_{x \rightarrow 0} \frac{\log x}{\operatorname{cosec} x}$
9. Evaluate $\lim_{x \rightarrow 0} \frac{\log(\tan x)}{\log x}$
10. Evaluate $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x) \tan x$
11. Evaluate $\lim_{x \rightarrow 0} x \log(\sin x)$
12. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$
13. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{\log(1+x)}{x^2} \right)$

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Assignment No.5 Topic: Expansion of Functions

1. Expand $\tan^{-1}(x)$ in powers of $(x - \pi/4)$.
2. Expand $\log(x)$ in powers of $(x - 1)$.
3. Expand $\sin(x)$ in powers of $(x - \pi/2)$.
4. Expand $3x^3 - 2x^2 + x - 4$ in powers of $(x + 2)$.
5. Expand $2x^3 + 3x^2 - 8x + 7$ in powers of $(x - 2)$.
6. Using Taylor's Series find $\sqrt{25.15}$.
7. Using Taylor's Series find $\sqrt{10}$.
8. Expand using Maclaurin's Series $\sin(x)$.
9. Expand using Maclaurin's Series $e^(-x)$.
10. Expand using Maclaurin's Series $\log(1 + x)$.

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Assignment No.6 Topic: Numerical Methods

1. Using Newton Raphson Method, find a root of $x^3 - 2x - 5 = 0$ correct up to 4 decimal places.
2. Using Newton Raphson Method, find a root of $x^4 - x - 9 = 0$ correct up to 4 decimal places.
3. Using Regula Falsi Method, find a root of $xe^x - 3 = 0$ correct up to 3 decimal places.
4. Using Regula Falsi Method, find a root of $xe^x - \sin x = 0$ correct up to 3 decimal places in $(-3, -2)$.
5. Solve the equation $10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14$, using Gauss Jacobi's Method in 5 steps.
6. Solve the equation $8x - y + 2z = 13, x - 10y + 3z = 17, 3x + 2y + 12z = 25$, using Gauss Jacobi's Method in 5 steps.
7. Solve the equation $20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25$, using Gauss Seidel's Method in 3 steps.
8. Solve the equation $5x - y = 9, -x + 5y - z = 4, -y + 5z = -6$, using Gauss Seidel's Method in 3 steps.

Assignment No.7 2413FEB1T1 : Engineering Mathematics-1

Topic: Complex Numbers

1. Find the argument and modulus of $z = -1 + i\sqrt{3}$.
2. If α and β are the roots of $z^2 \sin^2 \theta - z \sin 2\theta + 1 = 0$ then prove that $\alpha^n + \beta^n = 2 \cos n\theta \operatorname{cosec}^n \theta$.
3. Evaluate $z = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$ then find the value of $(z)^{10} + (\bar{z})^{10}$
4. Show that the n^{th} roots of the unity form Geometric Progression.
5. Prove that $(1 - \omega)^6 = -27$ where ω is a cube root of unity.
6. Find the roots of $x^4 + i = 0$.
7. Solve $X^6 - i = 0$.
8. If α is a root of $x^7 - 1$ then show that $(1 - \alpha)(1 - \alpha^2)(1 - \alpha^3)(1 - \alpha^4)(1 - \alpha^5)(1 - \alpha^6) = 7$.
9. Show that $\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$.
10. Expand $\cos^7 \theta$ in terms of multiple of θ .
11. Find $\cos^8 \theta + \sin^8 \theta$ in terms of multiples of θ .
12. Solve $\cosh x = 8 \sinh x = 1$ for real values of x .
13. If $\tanh x = \frac{1}{2}$ then find values of x and $\sinh 2x$.
14. If $u = \log \left[\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right]$ then prove that $\sinh u = \tan \theta$.
15. Prove that $\sinh^{-1}(\tan \theta) = \log(\sec \theta + \tan \theta)$.
16. Prove that $\sinh^{-1}(x) = \cosh^{-1} \sqrt{1 + x^2}$.
17. Separate into real and imaginary parts of $\sin(x + iy)$.
18. If $\sinh(\theta + i\phi) = e^{i\alpha}$ then prove that $\sinh^4 \theta = \cos^2 \alpha$.
19. Find the value of $\sin(\log i)$.
20. Separate into real and imaginary parts of $\log(3 + 4i)$.
21. If $e^{i\alpha} = i^\beta$ then prove that $\frac{\alpha}{\beta} = \frac{\pi}{2} + 2n\pi$.
22. Prove that $\log \left(\frac{a+ib}{a-ib} \right) = 2i \tan^{-1} \left(\frac{b}{a} \right)$.